

Study on Quantum Binomial Trees for Option Pricing

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Abstract Classical random walks on binomial trees are commonly used to model market fluctuations for calculating Option Prices. There is also interest in the use of Trinomial tree models, which can potentially improve computational efficiency. More recently, developments in Quantum Computing and the promise of potentially quadratic speedup has invited attempts to develop Quantum Walk algorithms for Option Pricing [7] [2]. However, a review of the published literature reveals no reliable demonstration of Quantum Walks implemented for option pricing calculations. This project studies the state of the art in the implementation of Quantum Walks on trees, and investigates the potential benefits of, and the challenges in implementing, Binomial Quantum Walk market fluctuation models for option pricing.

The code for this project can be found at <https://github.com/RajAnjali/quantum-finance>.

1 Introduction

The binomial and trinomial models are discrete-time lattice methods in computational finance for option pricing. These models approximate the stochastic evolution of the underlying asset price over time using a recombining tree structure. The binomial model was first introduced by Cox, Ross, and Rubinstein in 1979 [1]. The trinomial model extends binomial model by allowing three possible movements of the asset price at every time step instead of two. Both models provide intuitive probabilistic interpretations and are closely related to random walks and martingale theory [3].

The Binomial tree model has been extensively studied in the literature, but relatively little work exists on Trinomial tree models, in spite of potential computational benefits. These are therefore a suitable subject for studying the mathematical setup. The recombining structure of binomial and trinomial trees makes them attractive not only for classical numerical methods, but also for graph-based formulations of quantum walks. This motivates studying the relationship between classical lattice pricing methods and quantum walk algorithms, and assessing whether the theoretical quadratic speedups for quantum walks can be meaningfully applied to option pricing problems.

2 Trinomial Tree Model

This section develops the mathematical description of classical trinomial tree model that will later serve as the basis for discussing graph-based random walks and their quantum analogues.

2.1 Model Description

Just like binomial model, trinomial model also breaks down the time interval $[0, T]$ into finite number of time intervals, where T is maturity time of the option. The trinomial model incorporates three possible values that an underlying asset can have in one time period, where the asset price can move up, down or stay the same with certain probabilities [5]. If the jump sizes u, d and the transition probabilities p_u and p_d are known, it is possible to find the value of the underlying asset S , for any sequence of price movements. A directed graph with nodes labeled by asset prices and edges connecting nodes separated by one time step and a single price jump gives the desired binomial or trinomial pricing tree as seen in Fig 1.

Parameters and Notations

- $S(t)$: Value of asset price at time t .
- $B(t)$: Value of risk-free asset at time t .
- $V(t)$: Value of portfolio at time t .
- Δt : Length of each time interval.

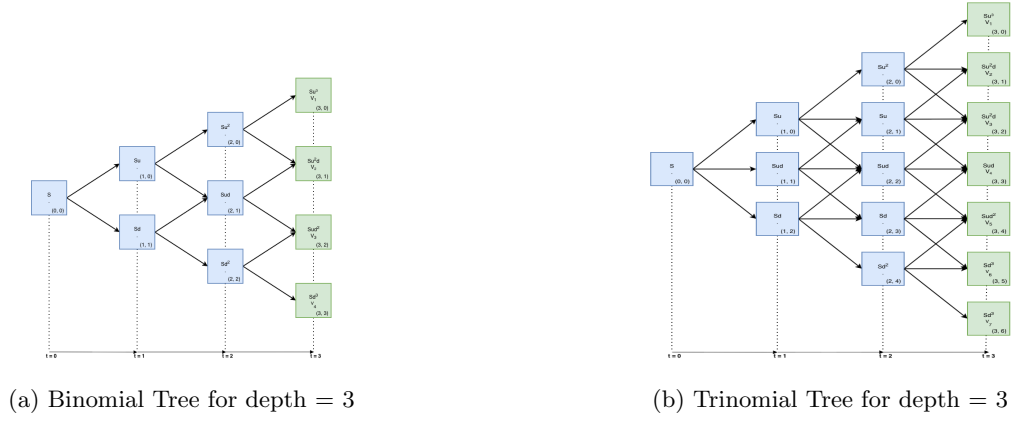


Figure 1: Caption for this figure with two images

- r : Risk-free rate of interest.
- σ : Volatility of stock price.
- f : Option price.
- u : rate of increase of stock price.
- d : rate of decrease of stock price.
- p_u : Probability of asset price increasing.
- p_m : Probability of asset price remaining same.
- p_d : Probability of asset price decreasing.

Assumptions

- The asset price can have only three possible outcomes on the following date i.e. price will either increase or decrease or remain same.
- Volatility σ of the asset and risk-free interest rate r are constant during the option time period.
- All time steps are equal.
- Risk neutral probabilities remain the same.
- $0 < d < 1 < u$.
- $ud = 1$, for computational simplicity.
- There are no arbitrage opportunities.
- Discounted value of stock is a martingale.
- Markets are efficient and complete.
- There is no friction in the market.

Probability Space The trinomial model is governed by the dynamics for the price $S(t)$ of the asset (stock):

$$S(t + \Delta t) = \begin{cases} S(t)u & \text{with probability } p_u \\ S(t) & \text{with probability } p_m \\ S(t)d & \text{with probability } p_d \end{cases}$$

Here $u > 1, d < 1, p_u, p_d \in (0, 1)$ and $p_m = 1 - p_u - p_d > 0$.

Let us define $\Omega_N = \{\omega = (y_1, y_2, \dots, y_N) : y_i \in \{+1, 0, -1\}, i \in \{1, 2, \dots, N\}\}$ for n-period trinomial model. Let F be the sigma algebra generated by Ω_N . Then (Ω_N, F, P) is the probability space with probability measure $P(\omega) = p_u^{N_u} p_m^{N_m} p_d^{N_d}$ where

$N_u = N_u(\omega)$ = Number of times stock price increases along a path ω

$N_m = N_m(\omega)$ = Number of times stock price remains same along a path ω

$N_d = N_d(\omega)$ = Number of times stock price decreases along a path ω

2.2 No arbitrage condition

The no-arbitrage condition guarantees that the resulting pricing model is financially consistent and admits a valid risk-neutral measure. This condition also ensures that the option pricing recursion remains stable across the tree. [6]

For one period trinomial model, we have $\Omega_1 = \{-1, 0, +1\}$,

$$P(\Omega_1) = 1 \implies p_u + p_m + p_d = 1 \quad (1)$$

From the martingale condition, we have

$$up_u + p_m + dp_d = e^r = a \quad (2)$$

On solving equations (1) and (2), we get

$$p_u = \left(\frac{a-d}{u-d} \right) - p_m \left(\frac{1-d}{u-d} \right) \quad (3)$$

$$p_d = \left(\frac{u-a}{u-d} \right) - p_m \left(\frac{u-1}{u-d} \right) \quad (4)$$

Note 2.2.1: $u > a \implies p_u + p_d < 1$

Note 2.2.2: $p_u > 0, p_d > 0, p_u + p_d < 1 \implies 0 < p_m < \left(\frac{u-a}{u-1} \right)$

On combining the results from note 2.2.1 and 2.2.2, we get no-arbitrage conditions as:

$$u > a, \quad 0 < p_m < \left(\frac{u-a}{u-1} \right) \quad (5)$$

Theorem 2.2.3: Under the condition (3), the portfolio will have no arbitrage opportunity.

Proof: Consider a portfolio $h_S(t), h_B(t)$, where $h_S(t)$ and $h_B(t)$ are positions in the stock and bond (risk-free asset) at time t . Let us assume there is an arbitrage opportunity.

At $t = 0$,

$$V(0) = h_S(0)S(0) + h_B(0)B(0)$$

At $t = 1$,

$$V(1) = h_S(0)S(0)u + h_B(0)B(0)e^r$$

$$V(1) = h_S(0)S(0) + h_B(0)B(0)e^r$$

$$V(1) = h_S(0)S(0)d + h_B(0)B(0)e^r$$

Since there exist an arbitrage opportunity, at $t = 0, V(0) = 0$ and $V(t) \geq 0$ for $t > 0$, where $V(t)$ is the value of the portfolio at time t .

$$\implies V(0) = h_S(0)S(0) + h_B(0)B(0) = 0$$

$$\implies h_S(0)S(0) = -h_B(0)B(0)$$

and

$$h_S(0)S(0)u + h_B(0)B(0)e^r \geq 0 \implies h_S(0)S(0)(u - e^r) \geq 0$$

$$h_S(0)S(0) + h_B(0)B(0)e^r \geq 0 \implies h_S(0)S(0)(d - e^r) \geq 0$$

$$h_S(0)S(0)d + h_B(0)B(0)e^r \geq 0 \implies h_S(0)S(0)(1 - e^r) \geq 0$$

Case 1: $h_S(0) > 0$

$$1 - e^r \geq 0 \implies e^r \leq 1 \implies r \leq 0$$

$$\text{But } r \geq 0, \quad \therefore r = 0$$

$$d - e^r \geq 0 \implies d > 1 \quad \text{which is a contradiction.}$$

Case 2: $h_S(0) < 0$

$$1 - e^r \leq 0 \implies e^r \geq 1 \implies r \geq 0$$

$$u - e^r \leq 0 \implies u < a \quad \text{which is a contradiction.}$$

Hence our assumption is wrong.

Therefore, there are no arbitrage opportunities if condition (3) is true. $\square$

2.3 Martingale Condition

This martingale condition forms the probabilistic foundation of risk-neutral pricing and connects the discrete trinomial model to the broader framework of modern mathematical finance.

The stock price $S(t)$ varies with t as:

$$S(t) = \begin{cases} S(t-1)u & \text{with probability } p_u \\ S(t-1) & \text{with probability } p_m \\ S(t-1)d & \text{with probability } p_d \end{cases}$$

Note 2.3.1: $S(t) = S_0 X(t)$ is a random variable on probability space (Ω, F, P) , where $X(0) = 0$ and $X(t) = \sum_{i=1}^t x_i$, $i \in \{1, \dots, t\}$, $\forall t \in 1, 2, 3, \dots$

$$x_i(\omega) = \begin{cases} u & \text{if } y_i = +1 \\ 1 & \text{if } y_i = 0 \\ d & \text{if } y_i = -1 \end{cases}$$

Note 2.3.2:

$$\begin{aligned} \mathbb{E}[S(t)] &= \mathbb{E}[S_0 \Pi x_i] = S_0 \Pi \mathbb{E}[x_i], \quad i \in \{1, \dots, t\} \\ &= S_0 \Pi (up_u + p_m + dp_d), \quad i \in \{1, \dots, t\} \\ &= S_0 (up_u + p_m + dp_d)^t \end{aligned}$$

Consider the discounted value of stock price $S(t)$, (i) $e^{-rt}S(t)$ is integrable

$$\mathbb{E}(|e^{-rt}S(t)|) = e^{-rt}\mathbb{E}(|S(t)|) < \infty$$

(ii) $e^{-rt}S(t)$ is adapted to filtration $S(t-1)$ (iii) $\mathbb{E}(e^{-rt}S(t)|S(t-1)) = e^{-rt}S(t-1)$ almost surely

$$\begin{aligned} \mathbb{E}(e^{-rt}S(t)|S(t-1)) &= e^{-rt}\mathbb{E}(S(t-1)x_t|S(t-1)), \quad i \in \{1, \dots, t\} \\ &= e^{-rt}S(t-1)\mathbb{E}[x_t] \\ &= e^{-rt}S(t-1)(up_u + p_m + dp_d) \end{aligned}$$

For $e^{-rt}S(t)$ to be a martingale, we require

$$\begin{aligned} e^{-rt}S(t-1)(up_u + p_m + dp_d) &= e^{-r(t-1)}S(t-1) \\ \implies up_u + p_m + dp_d &= e^r \end{aligned}$$

Theorem 2.3.3: The discounted value of stock is a martingale iff $up_u + p_m + dp_d = e^r$.

2.4 Option Pricing Formula

A risk-neutral world has two features that simplify the pricing of derivatives:

1. The expected return on a stock is the risk-free rate.
2. The discount rate used for the expected payoff on an option is the risk-free rate.

Risk neutral evaluation theory states that the value of the option price is equal to its expected payoff in a risk-neutral world discounted at the risk-free interest rate.

Suppose that the payoff from the option is f_d if the stock price goes down, f_u if the stock price goes up, and f_m if the price remains same. Then the expected future payoff from the option will $(p_u f_u + p_m f_m + p_d f_d)$ in a risk-neutral world, and hence the value of option today will be

$$V(f) = e^{-rT}(p_u f_u + p_m f_m + p_d f_d) \quad (6)$$

There exists a family of trinomial tree models for the obtained value of p_u and p_d . For pricing, we are going to consider the following representative of the family of equations: Jump sizes of stock price are

$$u = e^{+\sigma\sqrt{2\Delta t}}, \quad d = e^{-\sigma\sqrt{2\Delta t}}$$

And the transition probabilities are given by $p_u =$

$$p_u = \left(\frac{e^{\frac{r\Delta t}{2}} - e^{-\sigma\sqrt{\frac{\Delta t}{2}}}}{e^{\sigma\sqrt{\frac{\Delta t}{2}}} - e^{-\sigma\sqrt{\frac{\Delta t}{2}}}} \right)^2, \quad p_d = \left(\frac{e^{\sigma\sqrt{\frac{\Delta t}{2}}} - e^{-\frac{r\Delta t}{2}}}{e^{\sigma\sqrt{\frac{\Delta t}{2}}} - e^{-\sigma\sqrt{\frac{\Delta t}{2}}}} \right)^2, \quad p_m = 1 - p_u - p_d.$$

These recursive relations allow option values to be computed efficiently by backward induction from the terminal payoff values at maturity to the root of the tree.

Backward iterative formula for European option From equation (6), we can write the recursion formula for the value of European option as: $V_{(n,j)} = e^{-r\Delta t}(p_u V_{(n+1,j+1)} + p_m V_{(n+1,j)} + p_d V_{(n+1,j-1)})$ Option prices at leaf nodes of the are payoff of option.

$$\text{Payoff of call option} = \max(S(t) - K, 0)$$

$$\text{Payoff of put option} = \max(K - S(t), 0)$$

These payoff values will work as the base condition for the recursion.

Backward iterative formula for American option Since in American option, the option can be exercised before maturity also, so at each tree node we will have to compare the payoff from equation (6) with the payoff from early exercise and take the larger value as the value of option price at that node. So the recursion formula for the value of American option will be:

$$V_{(n,j)} = \max(S_{(n,j)} - K, e^{-r\Delta t}(p_u V_{(n+1,j+1)} + p_m V_{(n+1,j)} + p_d V_{(n+1,j-1)})) \quad (\text{Call option})$$

$$V_{(n,j)} = \max(K - S_{(n,j)}, e^{-r\Delta t}(p_u V_{(n+1,j+1)} + p_m V_{(n+1,j)} + p_d V_{(n+1,j-1)})) \quad (\text{Put option})$$

Here $V_{(n,j)}$ is the value of option at j^{th} node in n^{th} level of the stock price tree.

3 Complexity Analysis

The time and space complexity is quadratic in the depth of the tree since the number of stored nodes grows proportionally to the square of the tree depth.

Complexity analysis for d-depth trees				
Property	Binary Tree	Binomial Tree	Trinary Tree	Trinomial Tree
Nodes at level i	2^i	$i + 1$	3^i	$2i + 1$
Total vertices	$2^{d+1} - 1$	$\frac{(d+1)(d+2)}{2}$	$\frac{3^{d+1} - 1}{2}$	$(d + 1)^2$
Total edges	$2^{d+1} - 2$	$d(d + 1)$	$\frac{3^{d+1} - 3}{2}$	$3d^2$
Complexity	$O(2^d)$	$O(d^2)$	$O(3^d)$	$O(d^2)$

The complexity increases exponentially in case of full n-ary tree but only quadratically in case of binomial or trinomial tree models because of the recombining property, which makes large-scale pricing computations feasible.

4 Comparison of trinomial model with Binomial model

This project presented a recombining trinomial tree for options pricing with constant volatilities. This was done by assuming that the time steps and risk neutral probabilities remaining the same for the entire contract time. The results of trinomial and binomial model with that of Black-Scholes formula has been compared for different sizes of trees with the same set of parameters.

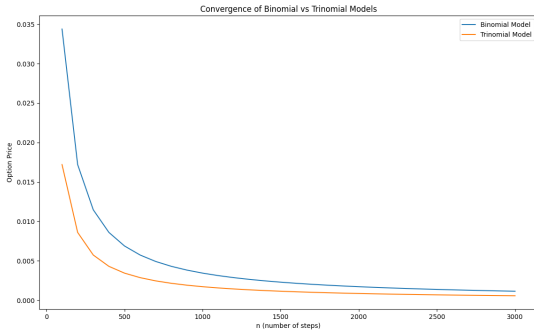


Figure 2: Error in binomial and trinomial model

model in terms of time complexity and accuracy.

Consider the pricing of an European call option with the following parameters: $S = 100.0, K = 100.0, T = 3, \sigma = 0.20, r = 0.05, \text{step} = 100, n_{max} = 3000$, Option type: European Call Option

Trinomial model has achieved accuracy up-to 3 decimal places in just 1800 step trinomial tree while binomial model could only achieve accuracy up-to 2 decimal places in 3000 step binomial tree. From the convergence plot, it can be deduced that the Trinomial method converges to the result of BlackScholes formula approximately two times faster than the Binomial model. Although n-step trinomial tree has more number of nodes than n-step binomial tree i.e. space complexity of trinomial tree is slightly larger but trinomial model performs better than binomial

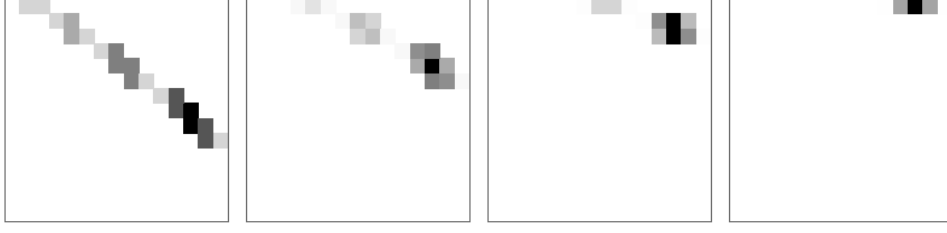


Figure 3: Transition matrix evolution for binomial tree of depth 4 as P^t , where $t = \{1, 2, 3, 4\}$

5 Random Walk on Binomial Tree

The recombining structure of binomial trees also allows them to be interpreted as graphs supporting random walk dynamics. Since quantum walks are generalizations of classical random walks, it is useful to examine the classical walk formulation before discussing quantum implementations.

The intuition for why a quantum walk can potentially achieve a quadratic speed-up can be visualized by the density plot of probability distribution of a binomial tree and its evolution after each walk step in figure (3).

Expected number of steps starting from leaf nodes to reach the root node in a d -depth binomial tree is quadratic in the depth of the tree. Therefore,

- Classical walk on binomial tree: $O(d^2)$
- Quantum walk on binomial tree: $O(d)$

The comparison highlights the primary motivation for studying quantum walks in financial applications. However, translating this theoretical advantage into practical option pricing algorithms remains a significant challenge.

6 Quantum circuit for Binomial Tree Model

Once the binomial tree is interpreted as a graph supporting a random walk, the next step is to encode the walk dynamics into a quantum circuit representation. The following construction shows how the dynamics of a binomial tree can, at least in principle, be encoded into a quantum circuit using standard quantum gates. However, the implementation complexity grows with tree depth, and practical pricing applications would additionally require efficient methods for encoding stock prices with high precision, computing payoff functions and extracting expectation values from the quantum state.

6.1 Quantum registers

- Coin register: 1 qubit
- Position i register: $\lceil \log(i) \rceil$ qubits
- Position j register: $\lceil \log(j) \rceil$ qubits

6.2 Initialization

The quantum walk starts from the leaf nodes, which means that i is the current level (depth) of the tree and j is all the nodes at that level. To initialize register j , we can apply Hadamard gate on each qubit in this register. To initialize register i , we have to apply X gate on qubits which represented the positions which have 1 in the binary representation of i .

6.3 One step of quantum walk

Coin toss operator C

Equivalent to applying Hadamard gate assuming the probability $p_u = p_d = 1/2$. This can be extended to arbitrary probabilities using appropriate rotation operators.

Position shift operator S

$$S |0\rangle |i\rangle |j\rangle = |0\rangle |i-1\rangle |j\rangle, \quad S |1\rangle |i\rangle |j\rangle = |1\rangle |i-1\rangle |j-1\rangle$$

The quantum walk operator for one step would be $U = S(C_c \otimes \mathbb{I}_i \otimes \mathbb{I}_j)$

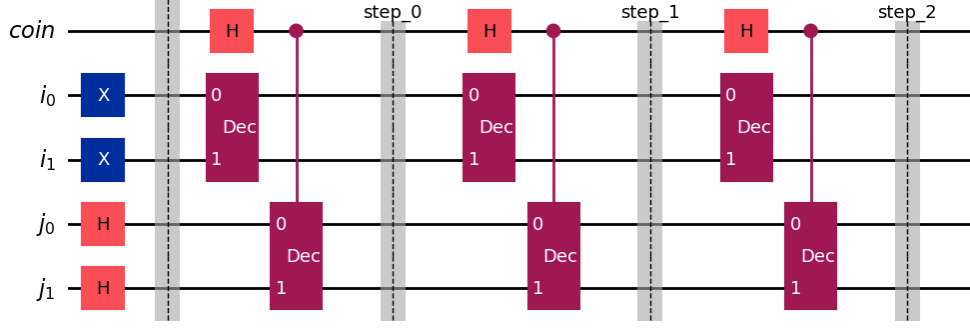


Figure 4: Quantum circuit encoding binomial tree

For binomial tree of depth 3, the i -th and j -th register needs 2 qubits each.

1. Start with the registers in $|0\rangle_c |00\rangle_i |00\rangle_j$
2. Initialize the registers for position i and j : $|0\rangle |11\rangle \sum_{x \in \{0,1\}^2} |x\rangle = \sum_{x \in \{0,1\}^2} |0\rangle |11\rangle |x\rangle$
3. Perform quantum walk step $d=3$ times (calculations shown for only one step below):
 - (a) Perform coin toss operation: $\sum_{x \in \{0,1\}^2} \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) |11\rangle |x\rangle = \frac{1}{\sqrt{2}} \sum_x |0\rangle |11\rangle |x\rangle + \frac{1}{\sqrt{2}} \sum_x |1\rangle |11\rangle |x\rangle$
 - (b) Perform shift operation: $\frac{1}{\sqrt{2}} \sum_x |0\rangle |10\rangle |x\rangle + \frac{1}{\sqrt{2}} \sum_x |1\rangle |10\rangle |x-1\rangle$

7 Challenges in quantum implementations

A major practical difficulty is that option pricing problems depend on large amounts of classical market data. Efficiently loading this information into quantum states is nontrivial and may offset theoretical speed advantages if implemented inefficiently. The architectures for implementing Quantum Random Access Memory (QRAM) is still a controversial issue, having concerns about its feasibility and fault tolerance [4].

Additionally, extracting the final option price from the quantum state requires repeated measurements and statistical estimation, introducing additional computational overhead. These considerations suggest that the theoretical complexity improvements of quantum walks do not automatically translate into end-to-end practical speedups.

8 Why variational quantum eigensolver is not immediately helpful

Variational quantum algorithms are a class of hybrid algorithms to solve optimization problems which uses quantum computers to compute cost function value and classical computers to iteratively update the parameters and minimize the total cost. Although the algorithms such as Variational Quantum Eigensolver (VQE) have shown promise in several optimization and quantum simulation problems, option pricing on lattice structures is fundamentally an expectation-estimation problem rather than a ground-state optimization problem. Therefore VQE does not naturally address the core computational bottlenecks in tree-based option pricing methods.

9 Conclusion

A review of the classical binomial and trinomial tree models has illustrated the challenges in implementation of quantum walks on trees and issues with existing claims. The use of both models has been demonstrated in python for the pricing of European and American options. The prices obtained using the Trinomial tree model have been shown numerically to converge to the Black-Scholes model faster than the Binomial tree model. A study of the graph structures used to represent binomial tree models and a rough sketch of theoretical time-complexities for quantum walks has also been presented, as well as techniques for the construction of quantum circuits for quantum walk on a binomial tree.

Overall, the project suggests that quantum walk formulations of lattice pricing models remain primarily theoretical at present. While the possibility of quadratic speedups is mathematically appealing, substantial challenges remain in state preparation, data encoding, and measurement efficiency. Nevertheless, the structural relationship between recombining trees and graph-based quantum walks provides an interesting direction for future research at the intersection of quantum computing and computational finance.

10 Future Research

The project can be extended to include the theoretical analysis of quantum query complexity using the Electrical Network Framework for quantum walks. Further research can also explore the mechanics of Quantum Amplitude Estimation (QAE) algorithm for option pricing [8], particularly to investigate if similar amplitude-based encoding of market data can be implemented for the quantum walk framework.

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